



Technical note

Investigating the genetic stock structure of blue marlin (*Makaira nigricans*) in the Pacific OceanSamuel M. Williams^{a,b,*}, Jamie Wyatt^a, Jennifer R. Ovenden^a^a Molecular Fisheries Laboratory, School of Biomedical Sciences, The University of Queensland, St Lucia, QLD 4072, Australia^b Department of Agriculture and Fisheries, Brisbane, QLD 4001, Australia

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ABSTRACT

We evaluated the population structure of *M. nigricans* within the Pacific Ocean using 17 microsatellite loci and 6204 single nucleotide polymorphisms. Evidence from both types of genetic markers supported the null hypothesis of a single genetic population throughout the Pacific Ocean. This study provides insight into the current stock structure of *M. nigricans* and discusses the implications for fisheries management.

1. Introduction

Many highly migratory species such as tunas, billfishes, sharks and rays are also important to commercial, recreational and artisanal fisheries. The istiophorid billfishes are known as one of the most highly mobile groups of fishes, with several species showing the capacity to move across ocean basins (Ortiz et al., 2003). Despite this capacity for dispersal, the influence of behavioral drivers such as reproductive philopatry have been shown to maintain population structure in some species (Domeier and Speare, 2012; Purcell and Edmands, 2011; Williams et al., 2015). This complex interplay between ocean migration and philopatry is not new in marine species, however, it presents challenges for defining biologically appropriate management units (Boustany et al., 2008; Domingues et al., 2018; Lascelles et al., 2014). To effectively assess and manage migratory fish stocks, the regional fisheries management organisations (RMFO) that govern the high seas fisheries, rely upon updated stock structure information to inform stock boundaries and cross jurisdictional management arrangements (Graves and McDowell, 2003; Hawkins et al., 2016). Whilst criteria based on morphological features, tag-capture and reproductive studies have been widely used to estimate stock structure in many highly migratory marine species, the use of genetic techniques have been shown to be effective when working over broad geographic distances and to test for reproductive philopatry (Graves and McDowell, 2003; Jorgensen et al., 2010; Kim and Sappington, 2013; Mamoozadeh et al., 2017).

The blue marlin (*Makaira nigricans*), an istiophorid billfish, is one of the world's largest species of fish and has a circum-global distribution

that ranges throughout temperate and tropical waters (Nakamura, 1985). The extent of understanding of *M. nigricans* stock structure throughout its range varies considerably among ocean basins (Graves and McDowell, 2015). In the Atlantic Ocean, extensive genetic data support the presence of connectivity throughout the basin (Buonaccorsi et al., 2001; Graves and McDowell, 2003, 2015). Comparisons of the genetic structure of *M. nigricans* among ocean basins have identified past periods of isolation and gene flow between the Atlantic and Pacific Oceans (Graves and McDowell, 1995; Graves, 1998). Although considerable effort has been undertaken to understand the connectivity of Atlantic Ocean stocks, there is still uncertainty around stock structure of *M. nigricans* in the other ocean basins (Graves and McDowell, 2015).

Within the Pacific Ocean, studies investigating *M. nigricans* genetic structure have indicated the presence of connectivity across the north Pacific (Chen et al., 2016). However, much of the previous work within the Pacific Ocean has focused on the use of mtDNA markers. The absence of powerful nuclear markers has constrained the capacity to conclusively resolve genetic structure within the basin (Chen et al., 2016; Williams et al., 2017). Due to these limitations, previous studies have highlighted the need for more sensitive nuclear markers in order to test for genetic structure in the Pacific Ocean for informing the assessment and management (Chen et al., 2016). Where nuclear markers have been applied to investigate genetic structure within the Pacific Ocean, the sample coverage has been isolated to the comparisons of locations in the north-eastern Pacific (Hawaii and California) (Buonaccorsi et al., 2001; Collette et al., 2006; Graves and McDowell, 2003).

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Given the capacity of *M. nigricans* to undertake large scale movements and the absence of physical barriers to movement across the Pacific basin, we hypothesize the absence of genetic population structure in *M. nigricans* across the Pacific Basin (Graves and McDowell, 2003; Williams et al., 2015). In this study, we test this hypothesis using a combination of nuclear microsatellite and single nucleotide polymorphism (SNP) markers to evaluate the contemporary genetic stock structure of *M. nigricans* across Pacific Ocean.

2. Methods

Muscle or fin tissue samples were obtained from *Makaira nigricans* located within various sites of three major regions in Pacific Basin: Central East Pacific, (Hawaii $n = 13$; Ecuador $n = 16$; and Mexico $n = 15$), Coral Sea (New Caledonia $n = 14$ and Eastern Australia; $n = 16$), and China Sea (Vietnam $n = 12$) (Supplemental Fig. S1). Tissue samples were obtained through non-lethal tag and release operations or from landed commercial product, with the co-operation of fisheries observers, recreational anglers and research collaborators. Samples were stored in either 20 % dimethyl sulphoxide salt solution (DMSO, 5 M NaCl) or 90 % ethanol at -20°C . DNA extraction was performed for all samples using the salting-out method as described by Sunnucks and Hales (1996). The quality and presence of high molecular DNA was confirmed using a 1% agarose gel electrophoresis.

Extracted DNA was sequenced and genotyped across 17 microsatellite loci using protocols for cross-specific amplification as outlined in Williams et al. (2014). Genotyping was performed using an Applied Biosystems® 3130 Genetic Analyzer with GeneMapper® v 3.7 Software. The presence of null alleles and potential scoring errors were assessed using MICRO-CHECKER v. 2.2.3 (Van Oosterhout et al., 2004). All loci were assessed for deviations from Hardy-Weinberg Equilibrium (HWE) and for the presence of linkage disequilibrium (LD) with Genepop 1.0.5 (Rousset, 2008). Microsatellite loci were deemed unsuitable for estimating population structure and were excluded from further analysis if they exhibited significant departures from HWE or LD in one or more locus-by-population comparison.

Twenty-seven samples yielded the required DNA both in terms of quality and quantity, enabling SNPs to be attained using DArTseq™ genome complexity reduction technology and double digest restriction-site sequencing (DD-RAD) method as described by Georges et al. (2018). Those samples were representative of two sampling regions (Central East Pacific: 17 samples; and Coral Sea: 10 samples). The DArTsoft14 proprietary software was used to assess quality, undertake read assembly and score SNPs based on technical sample replicates (i.e. samples processed twice from DNA library preparation to SNP calling) to optimise algorithm parameters (Williams, 2019). Markers and individuals were then filtered on various parameter thresholds to produce a final dataset of 6204 SNPs and 27 individuals in Radiator (Gosselin et al., 2020). These parameters included reproducibility, call rate, monomorphic and common markers, coverage, poor genotyping per locus and per individual, minor allele frequency (MAF), LD, heterozygosity and HWE (Supplemental Table 1). To avoid the possible influence of SNP clusters in downstream population structure analyses, the LD short filter in Radiator was used to identify any locus containing multiple SNPs and the exclusion criteria was set to retain only a single SNP per locus at random (Ahrens et al., 2018).

Genetic population analyses were performed separately on the filtered SNP and microsatellite loci datasets. For microsatellite markers, genetic indices including F_{IS} (inbreeding coefficient), observed and expected heterozygotes, and allelic richness were calculated using the package GENEPOP 1.0.5 (Rousset, 2008). For both SNP and microsatellite markers, population structure was estimated using F_{ST} values to measure the extent of divergence among populations relative to the net genetic diversity.

Cluster analysis was undertaken to determine the relationships among individuals from each region through discriminant analysis of

principle components (DAPC) with the package ADEGENET 2.1.1 (Jombart and Ahmed, 2011). Group membership for DAPC analysis was defined by region and Bayesian Information Criterion (BIC) scores were used to assess the optimal number of genetic clusters across a range of K values. Models with the lowest BIC scores were identified as representing the optimal number of K for use in DAPC analysis.

3. Results and discussion

A total of 16,491 SNPs were scored and filtered using the ‘filter_DArT’ pipeline in the Radiator package Gosselin, 2018. A total of 10,287 SNPs were unsuitable for downstream analyses as features such as call rate, monomorphism, coverage, MAF, LD, observed and expected heterozygosity, and HWE values diverged outside of the thresholds Supplemental Table S1. From the 14 filtering steps applied, the filters on call rate 3548 SNPs excluded, coverage 2721 SNPs excluded and LD 2395 SNPs excluded had the strongest influence when detecting non-conforming markers Supplemental Table S1. The final dataset after filtering included 6204 SNPs which were used for further evaluation of genetic population structure.

A final dataset of 16 microsatellite loci was established for evaluating population structure after locus ISIN017 was removed due to a missing data content of greater than 25 % across all individuals (Williams, 2019). The remaining loci were assessed for null alleles using Microchecker which suggested evidence for the presence of null alleles in loci ISIN001 and ISIN042 for samples from the Central East Pacific region (Van Oosterhout et al., 2004). In evaluation of the Microchecker results, the loci were checked for departures from HWE. Loci exhibited no evidence of deviation from HWE or presence of LD, with the exception of locus ISIN001 which showed significance for HWE deviation for Central East Pacific region. This deviation appeared due to a deficiency in heterozygotes and as this pattern was not evident across the remaining locations, locus ISIN001 was retained for downstream analysis of population structure. All loci were polymorphic but varied widely in the number of alleles (2–18) and heterozygosity (0.03–0.97). The most polymorphic locus was ISIN001 (mean number of alleles = 12.2), while the least polymorphic loci were ISIN011 and ISIN037 producing location averages of 1.8 and 2 alleles, respectively (Supplemental Table S2).

The level of genetic connectivity among sample locations was estimated for both the microsatellite and SNP datasets using pairwise F_{ST} comparisons. Evaluation of microsatellite pairwise F_{ST} estimates revealed no significant genetic differentiation among any locations within regions, as a result, these locations were pooled for subsequent analysis. Pairwise F_{ST} comparison between regions (Central East Pacific $n = 44$; China Sea $n = 12$; and the Coral Sea $n = 30$) were also not found to be significantly different from one another ($P > 0.05$). Microsatellite F_{ST} estimates ranged from -0.0038 to 0.0008 between regions (Table 1). For the SNP data, the F_{ST} value among samples from Central East Pacific and Coral Sea was also low (0.003) and non-significant ($P > 0.05$) (Table 1). The low F_{ST} value and non-significant pairwise comparisons among locations, suggests a lack of population structure between individuals from these regions within the Pacific Ocean.

These data also corroborated the findings of both DAPC for both

Table 1
Pairwise F_{ST} estimates for microsatellite loci and SNP loci. SNP comparisons are above the diagonal and microsatellite comparisons are displayed below the diagonal. There were no significant differences (< 0.05) among any of the pairwise comparisons.

	Coral Sea	Central East Pacific
Coral Sea		0.0030
Central East Pacific	0.0008	
China Sea	-0.0035	-0.0038

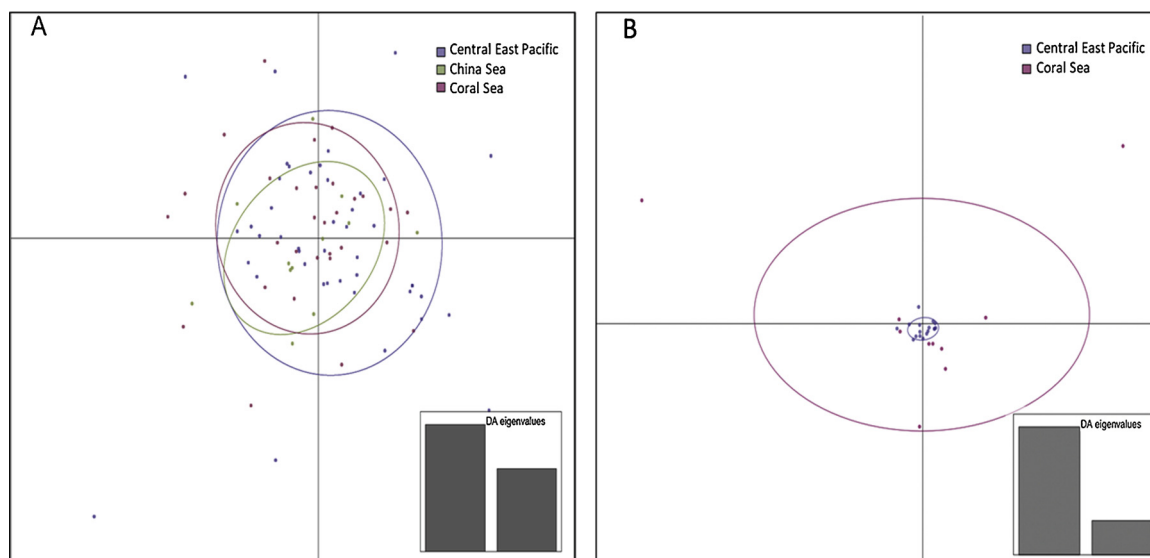


Fig. 1. Shows discriminate analysis of principle components plots applied to; (A) the microsatellite allele frequencies and; (B) SNP markers.

microsatellite and SNP datasets where the most likely number of clusters (K) for both datasets were equal to 1 (Supplementary material Figs. S2 & S3). The DAPC plot for microsatellite markers depicted an overall, even spread of genetic relatedness between individuals from Central East Pacific, China Sea and Coral Sea, thus suggesting an absence of genetic structuring (Fig. 1A & B). The DAPC analysis using SNP data also displayed genetic similarity among samples from each location, and therefore indicated a lack of population structure between the Central East Pacific and Coral Sea. Two outlier samples from the Coral Sea were observed within the DAPC analysis of SNP data. These samples were not identified to have non-representative biogeographic characteristics and demonstrated more consistent clustering within microsatellite analysis.

Our study was unable to detect genetic population structure for *M. nigricans* among the Pacific Ocean locations investigated. The presence of low (and non-significant) fixation indices and homogeneous clustering among locations indicate the presence of genetic connectivity within the Pacific Ocean. This evidence of genetic homogeneity is consistent with previous population genetic analysis on *M. nigricans* within the Pacific and Atlantic oceans (Buonaccorsi et al., 2001; Chen et al., 2016; Graves and McDowell, 2003). While we note limitation in the sample sizes included in this study, evaluation of the data through power analysis (see supplementary material) and support from existing tagging data provide additional evidence to uphold our null hypothesis that *M. nigricans* do not demonstrate genetic structure across the Pacific Basin (Ortiz et al., 2003). Due to the likelihood that statistical power may be low when sample sizes are low, we opted to not identify separate outlier and neutral SNP loci herein, however, we recommend that future studies be set up to include outlier analyses, and therefore assessment of adaptive differentiation among populations (Ahrens et al., 2018; Funk et al., 2012).

The presence of a single panmictic population of *M. nigricans* throughout the Pacific Ocean has previously been asserted after the observation of large-scale movements by several tagged animals. Conventional tagging of *M. nigricans* off Hawaii have demonstrated broad dispersal capabilities, with individuals traveling to waters off Mexico, Tahiti, Japan, and New Caledonia before being recaptured (Ortiz et al., 2003). This tag-recapture information is further supported by data from pop-up satellite tags, which have demonstrated movements of similar magnitude throughout the Central and Eastern Pacific Ocean (Carlisle et al., 2017).

Our findings are also consistent with the known reproductive biology of *M. nigricans*. Unlike other closely related billfishes such as *I.*

indica that have spatially restricted spawning and larval settlement areas (Domeier and Speare, 2012), the larvae of *M. nigricans* have been found extensively across tropical waters of the Pacific Ocean (30°S–20°N) (Nakamura, 1985; Pepperell, 2000). The extensive distribution of larvae along with the broad scale movement of adults provides a clear mechanism for gene flow throughout the Pacific basin. Given this, we suggest that the lack of genetic structure identified herein most likely indicates the presence of genetic mixing and provides evidence to support the assumption of a single *M. nigricans* stock within the Pacific basin.

The outcomes of this study further inform assessment and management of Pacific Ocean *M. nigricans*. The lack of spatial genetic differentiation supports the maintenance of a single ocean-specific management unit for *M. nigricans*. In addition, the successful implementation of nuclear SNPs and amplification of cross-species microsatellite markers in this study provide a baseline for future genetic investigations on *M. nigricans*. To further advance the understanding of *M. nigricans* additional population genetic studies are required throughout the Indian Ocean and to examine whether connectivity exists between the Indian and Pacific Ocean.

CRedit authorship contribution statement

Samuel M. Williams: Conceptualization, Methodology, Investigation, Formal analysis, Resources, Writing - original draft, Writing - review & editing. **Jamie Wyatt:** Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft. **Jennifer R. Oviden:** Conceptualization, Methodology, Resources, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.fishres.2020.105565>.

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